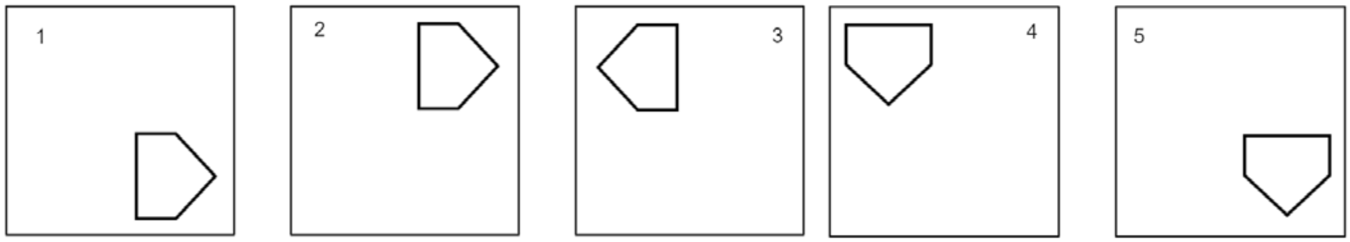
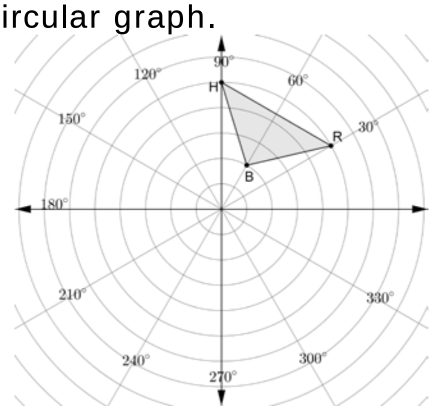
Geometry Name____________________
Unit 1 Date_____________________
Lesson 3 Homework Period____________________

Five frames that feature a polygon in different positions are shown.




| | Movement from… | Description |
|---|---|---|
| 1. | Frame 1 to Frame 2 | |
| 2. | Frame 2 to Frame 3 | |
| 3. | Frame 3 to Frame 4 | |
| 4. | Frame 4 to Frame 5 | |

Triangle $RHB$ is shown on the circular graph.




5. Plot each of the points.

| Point | Description |
|---|---|
| $W$ | Plotted on the $120°$ line and on the same circle as $B$. |
| $Y$ | Plotted on the $150°$ line and on the same circle as $R$. |
| $K$ | Plotted on the $300°$ line and on the same circle as $H$. |
| $T$ | Plotted on the $330°$ line and on the same circle as $B$. |
| $G$ | Plotted on the $360°$ line and on the same circle as $R$. |

6. Complete the statements about the triangles.

$\triangle HYW$ is a

- o reflection
- o rotation

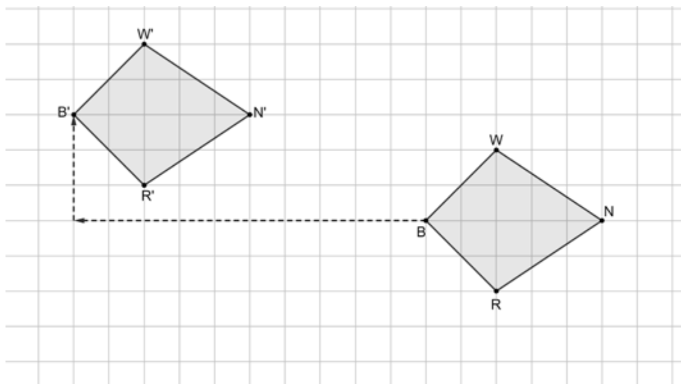
of $\triangle HRB$.

$\triangle GKT$ is a

- o reflection
- o rotation

of $\triangle HRB$.

A diagram of $WNRB$ and $W'N'R'B'$ is shown. Arrows are drawn from vertex $B$ to vertex $B'$.




Complete the statements.

7. The arrows show a possible path of movement. This path is called a

- o reflection.
- o rotation.
- o translation.

8. The same path

- o exists
- o does not exist

for all points on a diagram.

9. The path is called

- o isometry.
- o a mapping.

10. The transformation shown on the grid can be referred to as

- o an isometry
- o a mapping
- o a rigid motion

or as

- o an isometry
- o a mapping
- o a rigid motion

because the transformation

- o preserved distance.
- o preserved angle measure.
- o preserved angle measure and distance.